Introduction

Post-hoc domain generalization (**PDG**) methods deploy a final model *after* training using only source-domain information collected from an existing checkpoint bank. This makes PDG attractive in practice: the base model does not need to be retrained from scratch just to test a new deployment rule.

Strong PDG methods such as SWAD and DiWA already show that checkpoint averaging can be highly effective under DomainBed-style evaluation. But they primarily select deployments from *aggregate* signals:

- SWAD emphasizes **flatness** and overfit-aware averaging.
- DiWA emphasizes **diversity** across independently trained yet mergeable models.

The new paper narrative is narrower and cleaner:

- Aggregate source loss can hide which source domains are fragile.
- Diversity only helps when per-domain responses cancel in the right directions.
- **CDA therefore uses source-domain response geometry directly.**

Core poster claim: CDA selects and weights checkpoint soups using stability of *per-domain losses* under source-mixture shift, not just flatness or diversity alone.

Theoretical Motivation

Hidden-shift model. Let the source-domain loss vector be

$$L(\theta) = (L_1(\theta), \dots, L_E(\theta))^\top \in \mathbb{R}^E, \qquad \bar{L}(\theta) = \frac{1}{E} 1^\top L(\theta).$$

Assume the unseen target behaves like a nearby redistribution of source-domain mass:

$$L_T(\theta) \le \alpha_T^\top L(\theta) + \epsilon_{\text{app}}, \qquad \mathcal{A}_\rho = \{u_E + \delta : 1^\top \delta = 0,\ \|\delta\|_2 \le \rho$$

CDA target-risk certificate.

$$L_T(\theta) \le \bar{L}(\theta) + \rho \|P_\perp L(\theta)\|_2 + \epsilon_{\text{app}}.$$

Exact robust source risk.

$$\sup_{\alpha \in \mathcal{A}_\rho} \alpha^\top L(\theta) = \bar{L}(\theta) + \rho \|P_\perp L(\theta)\|_2 .$$

So $\bar{L}(\theta)$ alone is not enough. A model can have a low average source loss while remaining highly sensitive to source-mixture reweighting.

Why diversity is only useful directionally. For retained checkpoints with centered responses $c_j = P_\perp r_j$,

$$\|P_\perp z(w)\|_2^2 = \sum_{j=1}^{k} w_j^2 \|c_j\|_2^2 2 \sum_{1 \le i < j \le k} w_i w_j \langle c_i, c_j \rangle .$$

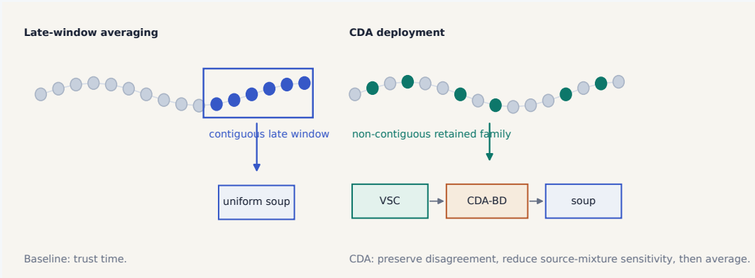
Diversity helps only when the cross terms $\langle c_i, c_j \rangle$ *cancel* hidden-shift sensitivity.

CDA Algorithm / Implementation

Input. A completed checkpoint bank

$$\mathcal{B} = \{\theta_1, \dots, \theta_K\} .$$

Output. One deployed soup $\bar{\theta}(w)$ together with source-side diagnostics.



Step 1: Version Space Compression (VSC)

For each source-support example (x_i, y_i), let $q_{j,i} = p_{\theta_j}(y_i \mid x_i)$. Define the full-bank disagreement mask

$$D_{\mathcal{B}}(i;\gamma) = 1\{\exists j : q_{j,i} \ge \gamma \text{ and } \exists j' : q_{j',i} < \gamma\} .$$

For a retained subset $S \subseteq \mathcal{B}$, the mismatch is

$$d_{\text{VSC}}(S, \mathcal{B}; \gamma) = \frac{1}{m} \sum_{i=1}^{m} 1\{D_S(i;\gamma) \ne D_{\mathcal{B}}(i;\gamma)\} .$$

VSC greedily retains a *non-contiguous* family that preserves the checkpoint bank's informative disagreement structure on source data.

Step 2: Nonuniform weighting with CDA-BD

Let $R \in \mathbb{R}^{E \times k}$ collect the retained family source-loss vectors and define the coordinatewise best target

$$b_e = \min_{1 \le j \le k} R_{e,j}.$$

CDA-BD chooses $w \in \Delta^k$ by solving

$$\min_{w \in \Delta^k} \frac{1}{E} \|Rw - b\|_2^2 \lambda_{\text{ent}} \mathrm{KL}(w \| u).$$

The deployed soup is

$$\bar{\theta}(w) = \sum_{j=1}^{k} w_j \theta_j.$$

Deployment bridge. The retained-family certificate for the final soup is

$$L_T(\bar{\theta}(w)) \le \bar{z}(w) \rho \|P_\perp z(w)\|_2 \left(E^{-1/2} + \rho\right) M(w) \epsilon_{\text{app}},$$

where $z(w) = \sum_j w_j L(\theta_j)$ and $M(w) = \left\|L(\bar{\theta}(w)) - z(w)\right\|_2$ is the *mergeability remainder*.

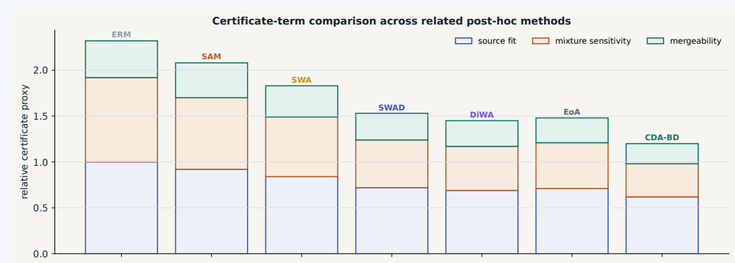
Results

Headline benchmark result. On the five DomainBed benchmarks, CDA reaches the strongest reported average OOD accuracy in the comparison package.

Method	PACS	VLCS	OH	TI	DN	Avg.
ERM	85.5	77.5	66.5	46.1	40.9	63.3
SWAD	88.1	79.1	70.6	50.0	46.5	66.9
DiWA	89.0	78.6	72.8	51.9	47.7	68.0
CDA	**89.1**	**79.5**	**72.9**	**52.3**	**47.8**	**68.3**

- CDA improves over SWAD by +1.4 average points.
- CDA improves over DiWA by +0.3 average points.
- CDA is the best row on 4/5 benchmark columns in this package.

Mechanism evidence. The paper's diagnostic layer measures the certificate terms directly rather than only reporting final accuracy.



Reading the diagnostic. SWAD mainly improves fit and mergeability through flatter deployment regions, DiWA reduces redundancy through diversity, and CDA is designed to reduce the *centered source-mixture sensitivity* term most directly.

Discussion

Takeaways

- Post-hoc DG is a **deployment problem**: choose a checkpoint family and a weight vector from source-only evidence.
- Flatness and diversity remain useful, but they are not the final deployment objective in CDA.
- CDA reframes checkpoint averaging around **distributional canalization**, i.e. stability of per-domain losses under source-mixture perturbations.

Limitations

- The theory assumes a **local source-supported hidden shift**. If the target is not well approximated by nearby source-mixture redistributions, the certificate can become loose.
- The margin over DiWA is **modest**. The main claim is not universal dominance; it is that a source-aware certificate yields a competitive and more interpretable deployment rule.

Selected references. [1] Gulrajani and Lopez-Paz, *In Search of Lost Domain Generalization*, ICLR 2021.
[2] Cha et al., *Domain Generalization by Stochastic Weight Averaging Densely*, NeurIPS 2021.
[3] Ramé et al., *Diverse Weight Averaging for Out-of-Distribution Generalization*, NeurIPS 2022.
[4] Arpit et al., *Ensemble of Averages*, NeurIPS 2022.